Priyansh Paliwal

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Education

VIT Bhopal University — Bhopal, India
Bachelor of Technology (B.Tech) — Sept 2023 – June 2027

- CGPA: 9.02

Technical Skills

Programming Languages: Java, Python, JavaScript, TypeScript
Frontend: HTML, CSS, React, Next.js, Tailwind CSS
Backend: RESTful APIs, FastAPI, Node.js, Prisma
AI & Machine Learning: RAG, Vector Search, ONNX
Database: PostgreSQL, ChromaDB, SQLite
Core Concepts: Object-Oriented Programming, DBMS, Data Structures and Algorithms, Computer Networks, Operating Systems, System Design
Tools & Platforms: Git, GitHub, Docker, Apache Kafka, Tree-Sitter, VS Code, Power BI

Projects

Real Estate Web App — Next.js, React, TypeScript, Tailwind CSS, PostgreSQL, Prisma
Demo | GitHub

- Built an end-to-end real estate portal using Next.js and Prisma ORM, featuring residential, commercial, and land listings backed by PostgreSQL.
- Engineered an interactive geospatial map experience using Leaflet and OpenStreetMap, enabling multi-property cluster visualization and a dynamic pin-drop coordinate picker for plot geolocation.
- Implemented role-based administrative workflows (RBAC) via NextAuth and secure Server Actions, enabling end-to-end property lifecycle management.
- Integrated Google Business Profile with the platform, driving real customer inquiries and reviews while improving local business discoverability.

Intelligent Codebase RAG Engine — Python, FastAPI, Next.js, Tree-sitter, ChromaDB
Demo | GitHub

- Architected a Hybrid GraphRAG engine for GitHub repositories, eliminating broken syntax scopes by parsing Tree-sitter ASTs to extract complete function boundaries and 480+ call/import dependencies into SQLite.
- Developed a zero-cost local embedding pipeline using CPU-accelerated ONNX (all-MiniLM-L6-v2) inside ChromaDB, cutting repository ingestion time to under 4 seconds with automated noise filtering.
- Reduced end-to-end query latency by 20x (from 19s to 0.9s) by benchmarking inference models and deploying Google Gemini 3.5 Flash-Lite with dual-mode zero-hallucination guardrails.
- Built an asynchronous FastAPI REST API that retrieves relevant codebase context and delivers grounded, cited explanations to a Next.js frontend in under 1 second.

Real-Time E-commerce Analytics Pipeline — Python, Docker, Apache Kafka, SQL, Power BI
GitHub

- Architected an event-driven streaming pipeline scaling horizontally across 3 Kafka partitions, achieving 18 ms average and 36 ms p95 event-processing latency.
- Implemented data validation and fault tolerance using Pydantic v2 and a Dead-Letter Queue (DLQ) to isolate malformed JSON and invalid transaction payloads without interrupting normal event processing.
- Containerized the streaming pipeline using Docker Compose, orchestrating Kafka, PostgreSQL, and 3 parallel consumer workers with persistent volumes for reproducible, cloud-deployable environments.
- Designed a 2-page real-time Power BI DirectQuery dashboard with 1–2 second Automatic Page Refresh (APR) for real-time KPIs including country revenue, latency percentiles and DLQ.

Certifications

- Top 1% (Elite + Gold, 99%), Marketing Analytics – **NPTEL** — Jan 2026 – Apr 2026
- Elite Certification, Introduction to Machine Learning – **NPTEL** — Jan 2025 – Apr 2025